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Adverse effects of GLP-1 receptor agonists: Clinical Implications, regulatory perspectives, and future directions

Riad Mohammed Abdelrahman [a,b,*], Taha Hussein Musa [c,d], Ismail Adam Arbab [c], Eltieb Omer Ahmed [e], Sahar Ibrahim Gasmallah [f], Mohammed Jalal [g], Chiamaka Linda Mgbechidinma [h], Wafaa Ramadan Ahmed [i,j]

[a] Clinical Pharmacy Department, Faculty of Clinical and Industrial Pharmacy, National University of Sudan, Khartoum, Sudan
[b] Faculty of Pharmacy, National Ribat University, Khartoum, Sudan
[c] Pharmacy Program, School of Medical & Health Sciences, Libyan International Medical University, Benghazi, Libya
[d] School of Medicine, Darfur University College, Nyala, Sudan
[e] Faculty of Pharmacy, Department of Pharmacy Practice, International University of Africa, Khartoum, Sudan
[f] College of Public and Environmental Health, University of Bahri, Khartoum, Sudan
[g] Faculty of Pharmacy, Karary University, Khartoum, Sudan
[h] School of Life Sciences, Centre for Cell and Development Biology and State Key Laboratory of Agrobiotechnology, The Chinese University of Hong Kong, Shatin, New Territories, Hong Kong, China
[i] Nursing Science Program. Libyan International Medical University, Benghazi, Libya
[j] Faculty of Nursing, Assiut University, Assiut, Egypt

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ABSTRACT

Background: Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have transformed the management of type 2 diabetes and obesity by enhancing glycemic control, facilitating weight reduction, and diminishing cardiovascular risk. But as more people use them, there is increased interest in understanding the side effects and monitoring them.

Objective: This review synthesizes current evidence about the spectrum, mechanisms, and regulatory responses to GLP-1RAs-associated adverse effects, integrating data from clinical trials, post-marketing surveillance, and pharmacovigilance reports.

Methods: A narrative review was conducted through a comprehensive search in Scopus, Web of Science, PubMed, and official regulatory databases (FDA, EMA) from 2006 to September 2025 using relevant terms. Key findings were organized by system involvement and regulatory milestones.

Results: Gastrointestinal events, including nausea, vomiting, diarrhea, and constipation, remain the most prevalent, typically transient and dose-dependent. Biliary and gallbladder disorders have been increasingly reported with long-acting agents and rapid weight loss. Post-marketing surveillance identified rare but serious events such as acute pancreatitis, intestinal obstruction, thyroid C-cell hyperplasia, acute kidney injury, and transient worsening of diabetic retinopathy. Cardiovascular outcome trials (LEADER, SUSTAIN-6, REWIND) demonstrated favorable safety, while recent reports (2023–2025) highlighted perioperative aspiration, neuropsychiatric events, and mild tachycardia, prompting updated labeling. Real-world pharmacovigilance data suggest

* Corresponding author. Faculty of Graduate Studies and Scientific Research, National Ribat University, Khartoum, Sudan.
E-mail addresses: riadsinnain@hotmail.com (R.M. Abdelrahman), taha.hm99@yahoo.com (T.H. Musa), ismail.adam@limu.edu.ly (I.A. Arbab), eltiebomer@gmail.com (E.O. Ahmed), Saharr.ib2014@gmail.com (S.I. Gasmallah), mohammedjalal@karary.edu.sd (M. Jalal), cmgbechidinma@gmail.com (C.L. Mgbechidinma), wafaa.ramadan@limu.edu.ly (W.R. Ahmed).

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that adverse event frequency varies by molecule, formulation, dose escalation schedule, comorbidities, and off-label use, underscoring the importance of individualized monitoring and risk mitigation strategies.
Conclusions: GLP-1RAs exhibit an overall favorable risk-benefit profile. Continuous post-marketing evaluation, risk mitigation strategies, and individualized therapy are vital as their therapeutic applications expand to obesity and metabolic disease beyond diabetes.

1. Introduction

1.1. Background and historical overview

The concept of gut peptides in glucose regulation dates back over a century, beginning with the discovery of secretin in 1902, and later supported by studies in 1964 showing the "incretin effect," where oral glucose provokes a stronger insulin response than intravenous glucose (Oehme and Wimmers, 1923; Edgard and Jean, 1929; On the treatment of Diabetus; ELRICK et al., 1964; Mcintyre et al., 1964; Creutzfeldt, 2005).

The first incretin hormone, gastric inhibitory polypeptide (GIP), was identified in the early 1970s, followed by glucagon-like peptide-1 (GLP-1) in the 1980s (Marathe et al., 2011). GLP-1 fulfills the criteria for an incretin hormone but differs from GIP in that its glucose-lowering effects are heavily influenced by delayed gastric emptying rather than solely insulinotropic or glucagonostatic actions (Marathe et al., 2011; Creutzfeldt, 1979; Mohd et al., 2025). In contrast, GIP has minimal impact on gastric emptying and may slightly accelerate it (Marathe et al., 2011).

In the 1990s, exendin-4, a GLP-1 receptor agonist (GLP-1RA) derived from Gila monster venom, was discovered, leading to FDA approval of exenatide for type 2 diabetes mellitus (T2DM) in 2005 (Nauck et al., 2021). Since then, GLP-1RAs have gained significant attention, with new analogs demonstrating clinical benefits (Ferhatbegović et al., 2023; Moll et al., 2024).

GLP-1, is a 30 or 31 amino acids gastrointestinal peptide hormone released by the intestinal tract, was first identified in 1984 (Abdelrahman et al., 2025). Since the introduction of exenatide in 2005, the class of GLP-1 receptor agonists has expanded to include liraglutide, dulaglutide, semaglutide, and, more recently, dual GIP/GLP-1 receptor agonists such as tirzepatide (Wang et al., 2023; Drug Approval Package).

In 2014, Liraglutide was approved as an adjunct therapy for long-term weight management in obese patients ≥12 years, based on BMI cutoffs (≥30 kg/m^2 in adults >60 kg). It was also authorized for the treatment of diabetes (Drug Approval Package), (Montanya and Sesti, 2009). Semaglutide followed in 2021 with the same weight management indication (FDA approves weight management

Table 1
FDA-approved GLP-1 RAs.

Trade name	Generic name	Population (indication)	Approval year	Key adverse effects
Byetta	Exenatide	Type 2 diabetes	2005	Nausea, vomiting, diarrhea, injection-site reactions, pancreatitis, hypoglycemia (with sulfonylureas/insulin)
Victoza	Liraglutide	Type 2 diabetes	2010	Nausea, diarrhea, vomiting, headache, pancreatitis, gallbladder disease, thyroid C-cell tumors (boxed warning)
Trulicity	Dulaglutide	Type 2 diabetes	2014	Nausea, vomiting, diarrhea, abdominal pain, decreased appetite, pancreatitis, injection-site reactions
Saxenda	Liraglutide	Obesity/ overweight	2014	Nausea, vomiting, diarrhea, constipation, pancreatitis, gallbladder disease, thyroid C-cell tumors (boxed warning)
Adlyxin	Lixisenatide	Type 2 diabetes	2016	Nausea, vomiting, headache, dizziness, hypoglycemia (with insulin/ sulfonylureas), pancreatitis
Xultophy	Liraglutide + Insulin Degludec	Type 2 diabetes	2016	Hypoglycemia, weight gain, nausea, diarrhea, injection-site reactions, pancreatitis, thyroid C-cell tumors
Soliqua	Lixisenatide + Insulin Glargine	Type 2 diabetes	2016	Hypoglycemia, nausea, diarrhea, headache, weight gain, pancreatitis
Bydureon BCise	Exenatide	Type 2 diabetes	2017	Nausea, diarrhea, injection-site nodules/reactions, pancreatitis, hypoglycemia (with insulin/sulfonylureas)
Ozempic	Semaglutide	Type 2 diabetes	2017	Nausea, vomiting, diarrhea, constipation, abdominal pain, gallbladder disease, thyroid C-cell tumors (boxed warning), rare pancreatitis
Rybelsus	Semaglutide (oral)	Type 2 diabetes	2019	Nausea, diarrhea, abdominal pain, decreased appetite, vomiting, rare pancreatitis, gallbladder disease
Wegovy	Semaglutide	Obesity/ overweight	2021	Nausea, vomiting, diarrhea, constipation, abdominal pain, headache, gallbladder disease, thyroid C-cell tumors (boxed warning), rare pancreatitis
Mounjaro	Tirzepatide	Type 2 diabetes	2022	Nausea, diarrhea, vomiting, decreased appetite, constipation, abdominal pain, gallbladder disease, possible thyroid C-cell tumors, pancreatitis
Zepbound	Tirzepatide	Obesity/ overweight	2023	Nausea, vomiting, diarrhea, constipation, abdominal pain, gallbladder disease, pancreatitis, possible thyroid C-cell tumors

Source: https://www.fda.gov/drugs/drug-safety-and-availability/update-fdas-ongoing-evaluation-reports-suicidal-thoughts-or-actions-patients-taking-certain-type

drug).

Prescriptions of GLP-1RAs rose by 300% between 2018 and 2023, reflecting their efficacy, safety profile, and convenient dosing. However, their high cost raises concerns regarding accessibility and potential impacts on healthcare equity (Rx History).

Lixisenatide, approved in 2016 for adults with type 2 diabetes as a once-daily adjunct to diet and exercise, demonstrated beneficial effects on weight reduction and was generally well tolerated. However, it has not been authorized for obesity treatment (Leon et al., 2017).

Semaglutide received multiple approvals: for type 2 diabetes in 2017, to reduce major cardiovascular events in patients with diabetes and cardiovascular disease in 2020, and to lower risks of kidney disease progression and cardiovascular death in patients with diabetes and chronic kidney disease in 2024 (Ozempic).

In 2021, it was also approved for chronic weight management in overweight or obese adults with at least one weight-related comorbidity (e.g., hypertension, type 2 diabetes, or dyslipidemia). Administration is via weekly subcutaneous injection. While both oral and injectable semaglutide are FDA-approved for diabetes, only the injectable form is authorized for obesity (Knudsen and Lau, 2019; Christou et al., 2019; Cowart, 2020; FDA Approves New Drug Treatment).

Dulaglutide, another long-acting GLP-1RA, was FDA approved in 2014 for type 2 diabetes. Higher doses have been shown to promote weight reduction (Gerstein et al., 2019; Østergaard et al., 2016; Jendle et al., 2016). However, it is not approved for obesity (Trulicity). Tirzepatide was approved for diabetes in 2022 and subsequently for obesity in 2023 (Update on FDA).

To date, only Liraglutide, semaglutide, and Tirzepatide have FDA approval for obesity/overweight treatment (Update on FDA).

Emerging GLP-1/GIP dual agonist, GLP-1/glucagon co-agonist agents currently progressing through clinical trials-such as Orforglipron, Mazdutide, Retatrutide, Ecnoglutide, and Survodutide-are poised to further broaden the therapeutic landscape of GLP-1 receptor agonists. Concurrently, the introduction of generic Liraglutide and evolving insurance coverage policies are transforming access to and affordability of these therapies (Monti et al., 2022). A summary of FDA-approved GLP-1RAs is depicted in Table 1.

1.2. Expanding indications and global use

Recent advances in GLP-1R agonist (GLP-1RA) therapy have transformed diabetes and obesity management (Chiquette et al., 2026; Rao, 2026). The therapeutic success of GLP-1RAs has driven exponential global adoption, supported by robust cardiovascular outcomes trials such as LEADER (Liraglutide), SUSTAIN-6 (semaglutide), and REWIND (Dulaglutide) (Bethel et al., 2018). Beyond glycemic control, pleiotropic effects such as weight loss, lipid improvement, and blood pressure reduction have driven their expanding role in obesity and cardiometabolic prevention.

However, the surge in prescribing has been accompanied by a parallel rise in adverse event reporting, including gastrointestinal intolerance, pancreatitis, gallbladder disorders, thyroid concerns, and emerging psychiatric effects (Drucker, 2024). The increasing off-label use of GLP-1RAs for cosmetic weight loss has further intensified scrutiny of their safety profile (Semaglutide Patient Safety Updates; Spitery et al., 2024).

Evidence consistently shows that GLP-1RAs are generally safe and effective in producing moderate weight loss (Narayan et al., 2025; Menon et al., 2024; Kramer et al., 2024; Liu et al., 2025). Their off-label use for weight reduction has become widespread, sometimes limiting access for type 2 diabetes patients who need them most (Powell and Taylor, 2024). Still, concerns about their safety have been raised in several reports (Ryan et al., 2021; Zeng et al., 2023; Wu et al., 2022).

Despite these advantages, GLP-1RAs are associated with a spectrum of adverse effects, most commonly gastrointestinal disturbances, and rare but serious risks affecting the pancreas, thyroid, and cardiovascular system (Adverse effects of GLP; Suissa et al., 2024), (FDA's Concerns with Unapproved GLP). Also, within a year of GLP1RA discontinuation, 49 % patients were heavier than they were when they started GLP1RAs (Abdel-Bary et al., 2025).Understanding these safety considerations is crucial for optimizing therapy, improving patient adherence, and mitigating potential complications.

1.3. Purpose of this review

While several systematic reviews and meta-analyses have addressed specific adverse effects, a comprehensive narrative synthesis that integrates the timeline of detection, mechanistic insights, clinical consequences, pharmaceutical responses, and future safety strategies remains lacking. As these agents continue to evolve into multifunctional metabolic drugs, a detailed appraisal of their adverse event profile is critical for clinicians, regulators, and policymakers alike.

A comprehensive understanding of global research on GLP-1RA-related adverse effects is essential to evaluate the current evidence base and to guide future investigations, especially in light of the expanding clinical use of these agents in diabetes, obesity, and metabolic disorders.

So, this narrative review on the adverse effects of GLP-1RAs is timely and valuable, given their rapid clinical uptake and ongoing safety debates. It aims to trace the historical evolution of GLP-1RA-related safety concerns, synthesize evidence on their adverse effects and underlying mechanisms, explore the impact on clinical practice and pharmaceutical industry dynamics, and outline strategies and research priorities for improving patient safety.

2. Methods

Data were collected to explore global research on GLP-1RA-related adverse effects. The search strategy included the keywords "GLP-1RA," specific trade names of individual agents, and terms such as "adverse effect," "side effect," "adverse event," and "safety."

Searches were conducted across Scopus, Web of Science, PubMed, and official regulatory authority websites from 2006 to September 2025 to ensure comprehensive coverage of research on GLP-1RA-related adverse effects. This period captures the complete publication history relevant to post-marketing safety observations and emerging adverse effect profiles up to the most recent year available in the different databases.

3. Timeline of detection and regulatory milestones

The first GLP-1 receptor agonist, exenatide, was approved by the FDA in 2005 for the treatment of type 2 diabetes. Shortly after approval, spontaneous reports emerged in the FDA Adverse Event Reporting System (FAERS) indicating nausea, vomiting, and pancreatitis. Between 2007 and 2011, several case reports, animal studies, and preclinical investigations suggested potential concerns regarding C-cell hyperplasia and pancreatic inflammation, raising regulatory attention (Thomsen et al., 2025).

Regulatory agencies, including the FDA and EMA, reviewed these data and concluded that while a direct causal link to pancreatitis could not be definitively established, continued post-marketing surveillance and warnings were warranted (Committee for Medicinal Products for Human Use (CHMP), 2014). In response, pharmaceutical companies such as Novo Nordisk and Eli Lilly implemented enhanced risk management strategies, including patient education programs, labeling updates, and recommendations for early recognition of abdominal symptoms. These actions helped establish a framework for monitoring adverse events in post-marketing settings.

From 2013 onwards, cardiovascular safety became a key regulatory requirement for glucose-lowering agents. Large-scale outcome trials, including LEADER (Liraglutide, 2016), SUSTAIN-6 (Semaglutide, 2016), and REWIND (Dulaglutide, 2019), demonstrated not only cardiovascular safety but in some instances superior cardiovascular outcomes, alleviating early concerns about cardiovascular risk.

Since 2020, the introduction and widespread adoption of Semaglutide and Tirzepatide-including their approval for obesity management (Wegovy®, Zepbound ®)-have significantly expanded the GLP-1RA market. This rapid increase in use has been accompanied by drug shortages, increased online promotion, and off-label or unsupervised use, raising new safety concerns (Semaglutide Patient Safety Updates).

Recent pharmacovigilance alerts (FDA, 2023; EMA, 2024) have highlighted emerging risks such as intestinal obstruction and suicidal ideation, based on post-marketing reports (EMA). Ongoing investigations aim to determine causality and identify patient subgroups at higher risk.

Overall, regulatory evaluation of GLP-1RAs has shifted from initial organ-specific concerns to a more comprehensive assessment of long-term systemic safety and population-level effects. A chronological overview of major GLP-1RA safety events and corresponding regulatory actions is summarized in Fig. 1.

4. The known and emerging adverse effects associated with GLP-1 receptor agonists

Safety data from randomized controlled trials, cardiovascular outcome studies, and post-marketing surveillance indicate that the adverse-effect profile of GLP-1 receptor agonists (GLP-1RAs) is largely mechanism-driven, dose-dependent, and consistent across agents. While gastrointestinal intolerance dominates early treatment phases, serious or irreversible adverse outcomes remain uncommon, and most safety signals have been clarified or attenuated through long-term trial evidence and real-world data.

4.1. Gastrointestinal adverse effects

Gastrointestinal (GI) adverse effects-including nausea, vomiting, diarrhea, constipation, and reflux-are the most frequently reported events associated with GLP-1RA therapy, occurring in approximately 40–60% of patients, particularly during treatment initiation and dose escalation (Adverse effects of GLP; Ismail et al., 2025; Sodhi et al., 2023; Chiang et al., 2025). Across early development programs for exenatide and liraglutide, nausea emerged as the principal dose-limiting factor, typically peaking within the first weeks of therapy and declining with continued exposure.

Trial data consistently demonstrate that GI intolerance is influenced by pharmacokinetics and dose intensity. The LEAD-6 trial,

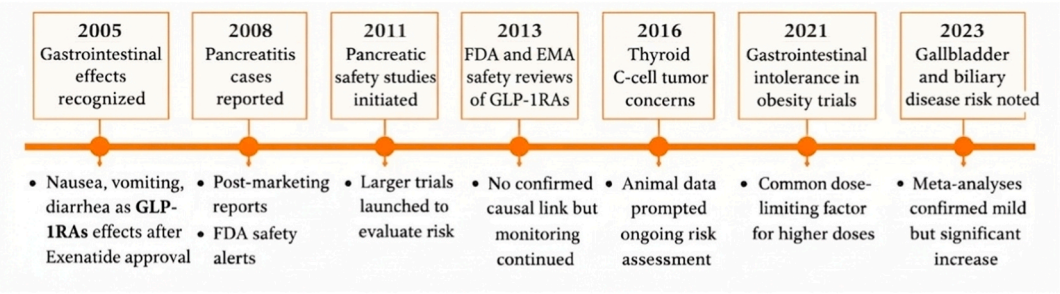


Fig. 1. Chronological overview of major GLP-1RA safety events and regulatory actions (2005-2025).

revealed that Liraglutide and exenatide were well tolerated, but nausea was less persistent (estimated treatment rate ratio 0.448, $p < 0.0001$) and minor hypoglycemia less frequent with liraglutide than with exenatide (1.93 vs 2.60 events per patient per year; rate ratio 0.55; 95% CI 0.34 to 0.88; $p = 0.0131$; 25.5%vs 33.6% had minor hypoglycemia) (Madsbad, 2009). Similarly, cardiovascular outcome trials such as LEADER, REWIND, and SUSTAIN-6 confirmed that while GI symptoms were common, they were rarely severe and infrequently resulted in serious complications (Marso et al., 2016a; Marso et al., 2016b; Thethi et al., 2021).

Higher-dose regimens used in obesity management amplify this effect. The STEP trials of semaglutide 2.4 mg demonstrated markedly higher rates of nausea and vomiting, with GI adverse events being the leading cause of treatment discontinuation compared with placebo (Wilding et al., 2021). These findings underscore the balance between enhanced weight-loss efficacy and tolerability at supratherapeutic doses.

Beyond transient symptoms, delayed gastric emptying has raised concerns regarding gastroparesis and intestinal obstruction, particularly in perioperative settings (Drucker, 2024; Kim and Yoo, 2025). Although randomized trials have not established a causal relationship, pharmacovigilance reports and mechanistic plausibility have led anesthesiology societies to recommend withholding GLP-1RAs prior to elective surgery to reduce aspiration risk (Thomsen et al., 2025).The FDA has reinforced these warnings while emphasizing that severe GI complications remain rare (New FDA warning label for; Drug Safety).

Severe GI events, including intestinal obstruction and gastroparesis, have been reported sporadically in pharmacovigilance databases such as WHO VigiBase, although randomized trials have not established causality (Adverse effects of GLP; Ismail et al., 2025; EMA, 2014; Faillie et al., 2022; Gudin et al., 2020; Nielsen et al., 2025). These reports have prompted increased regulatory scrutiny without altering the overall benefit–risk assessment.

Early post-marketing reports of acute pancreatitis following exenatide use raised concerns regarding pancreatic safety (Adverse effects of GLP; Ismail et al., 2025; Kim and Yoo, 2025). However, large randomized trials and meta-analyses-including LEADER, ELIXA, EXSCEL, and REWIND-have not demonstrated a statistically significant increase in pancreatitis or pancreatic cancer risk (Karagiannis et al., 2023; Nreu et al., 2023; Li et al., 2014; Thomsen et al., 2015). Some registry-based studies have even suggested a reduced incidence of obesity-related malignancies, including pancreatic cancer, potentially reflecting metabolic benefits rather than drug-specific effects (Wang et al., 2024).

Data remain less consistent for higher-dose regimens used in obesity management, likely due to shorter follow-up durations and heterogeneity in study populations (Sodhi et al., 2023; Rodriguez et al., 2024; Lomeli et al., 2024). Current consensus supports discontinuation of GLP-1RAs if pancreatitis is suspected and avoidance of rechallenge in affected individuals.

Gallbladder and biliary disorders have been reported across several trials, particularly those associated with rapid weight loss, such as STEP and AWARD. These events are thought to result from altered bile acid metabolism and reduced gallbladder motility rather than direct drug toxicity, warranting monitoring in patients with pre-existing biliary disease (Adverse effects of GLP; Ismail et al., 2025).

4.2. Hypoglycemia risk

GLP-1RAs are associated with a low intrinsic risk of hypoglycemia due to their glucose-dependent stimulation of insulin secretion. Across controlled trials, hypoglycemia rates ranged from 10.4% to 19.4%, with the majority of events occurring in patients receiving concomitant insulin or sulfonylureas (Adverse effects of GLP; Ismail et al., 2025; Ja'arah et al., 2021; Zhao et al., 2021). Importantly, cardiovascular outcome trials such as LEADER, EXSCEL, and REWIND confirmed that severe hypoglycemia was uncommon and occurred substantially less frequently than with traditional insulin secretagogues.

Earlier concerns regarding hypoglycemia in normoglycemic individuals have not been substantiated by real-world evidence, which consistently demonstrates a favorable safety profile when GLP-1RAs are used as monotherapy or with appropriate dose adjustment of background therapies (Marso et al., 2016a; Marso et al., 2016b). Ongoing FDA surveillance continues to monitor hypoglycemia risk across marketed agents, particularly as indications expand beyond diabetes (Ja'arah et al., 2021).

4.3. Thyroid and renal considerations

Rodent studies demonstrating C-cell hyperplasia and medullary thyroid carcinoma (MTC) at supratherapeutic exposures led to FDA-mandated boxed warnings for all GLP-1RAs. However, long-term human data from cardiovascular outcome trials and large observational cohorts have not confirmed an increased risk of thyroid malignancy (FDA's Concerns with Unapproved GLP; Gier et al., 2012). As a precaution, GLP-1RAs remain contraindicated in individuals with MEN2 or a personal or family history of MTC.

Acute kidney injury is a rare complication, most commonly secondary to volume depletion from severe GI adverse effects rather than intrinsic nephrotoxicity. Renal function monitoring is recommended in older adults and during periods of significant nausea or vomiting (Badve et al., 2025).

4.4. Cardiovascular events

Cardiovascular outcome trials consistently demonstrate a 12–15% reduction in major adverse cardiovascular events (MACE) with GLP-1RA therapy, despite a mild, dose-dependent increase in resting heart rate of 2–4 beats per minute (Badve et al., 2025). This chronotropic effect has not translated into increased arrhythmia-related morbidity or mortality.

Principal findings of the LEADER demonstrated that Liraglutide reduced major adverse cardiovascular events compared with placebo (13.0% vs 14.9%; HR 0.87, 95% CI 0.78–0.97), with significant reductions in cardiovascular death (4.7% vs 6.0%) and

myocardial infarction (6.3% vs 7.3%), while stroke rates were not significantly different (3.7% vs 4.3%). (Marso et al., 2016a; Marso et al., 2016b; Mentz et al., 2018; Thethi et al., 2021).

The most recent large-scale study, SELECT (Semaglutide Effects on Heart Disease in Overweight or Obese People without Diabetes, 2023), confirmed a cardiovascular benefit of semaglutide while noting GI intolerance and biliary events as the most frequent adverse effects. The FDA approved cardiovascular indications for Semaglutide in 2024 (Lincoff et al., 2023).

4.5. Suicidal ideation and psychiatric effects

Post-marketing reports have described mood changes, anxiety, and suicidal ideation during GLP-1RA therapy (FDA's Concerns with Unapproved GLP; EMA), ("New FDA warning label for), (Manne-Goehler and Franco, 2025; Ruggiero, 2024; Carmellini et al., 2025; Zheng et al., 2024; Moiz et al., 2025). While randomized trials have not demonstrated a causal association, regulatory agencies continue active surveillance. Proposed mechanisms include central GLP-1 receptor activity affecting mood regulation and hypothalamic–pituitary–adrenal axis signaling. Clinical vigilance is recommended in patients with pre-existing psychiatric conditions.

4.6. Other adverse effects

Persistence with GLP-1RAs is significantly lower in real-world settings (Karagiannis et al., 2023). Discontinuation is mainly driven by gastrointestinal adverse effects, with additional contributions from cost, access, and injection burden-particularly at higher doses used in obesity (e.g., STEP trials) (Wilding et al., 2021). Socioeconomic disparities further affect treatment uptake and continuation (Karagiannis et al., 2023; Lassen et al., 2024).

Overall safety is well supported, with no confirmed increased risk of pancreatic or thyroid malignancy (Lassen et al., 2024); however, GI intolerance remains the leading cause of discontinuation, and gallbladder-related events have been reported, especially with rapid weight loss (Zheng et al., 2023).

In the SUSTAIN-6 trial, semaglutide was associated with a transient increase in diabetic retinopathy complications, likely related to rapid glycemic improvement in patients with pre-existing disease; this signal has not been consistently replicated in subsequent trials (Zeng et al., 2023; Zheng et al., 2023; Vilsbøll et al., 2018). Gradual glycemic control remains advisable in high-risk individuals.

Additional rare events, including injection-site reactions, cutaneous hypersensitivity, and reversible intestinal obstruction, have been reported but typically resolve after treatment discontinuation (Jones and Cappola, 2025; Gao et al., 2025). Given the widespread use of GLP-1RAs across obesity, cardiovascular prevention, and reproductive-age populations, ongoing pharmacovigilance remains essential.

Notably, GLP-1RAs may restore ovulation and increase fertility in women with metabolic disorders, particularly polycystic ovary syndrome, increasing the risk of unintended pregnancy. Because these agents are contraindicated during pregnancy, counseling regarding contraception and treatment discontinuation before conception is recommended (Couldwell et al., 2026).

Table 2 summarizes the GLP-1RA Adverse Effects by System, Mechanism, and Clinical Relevance.

Table 2
Summary of GLP-1RA adverse effects by system, mechanism, and clinical relevance.

System/Organ	Adverse Effect(s)	Proposed Mechanism(s)	Clinical Relevance/Management
Gastrointestinal	Nausea, vomiting, diarrhea, constipation, abdominal discomfort	Delayed gastric emptying; activation of central GLP-1 receptors in brainstem leading to decreased gastric motility and increased satiety	**Most frequent AE class; dose-dependent and transient.** Usually improves with gradual dose escalation; symptomatic management if persistent.
Pancreatic	Pancreatitis (acute, rare)	Exocrine pancreatic stimulation; ductal hypertrophy and increased enzyme secretion (theoretical)	**Low absolute risk**; post-marketing surveillance ongoing. Discontinue immediately if suspected.
Hepatobiliary/ Gallbladder	Cholelithiasis, cholecystitis, biliary colic	Rapid weight loss altering bile composition; reduced gallbladder motility due to GLP-1 receptor–mediated effects	**Occasional (1–2%)**; monitor patients with rapid weight reduction or gallstone history.
Thyroid	C-cell hyperplasia, medullary thyroid carcinoma (in rodents)	GLP-1 receptor stimulation of parafollicular C-cells (observed in rodents, not humans)	**No confirmed human risk.** Class-label boxed warning remains (FDA). Contraindicated in patients with MEN-2 or MTC history.
Ocular/Retinal	Diabetic retinopathy worsening (Semaglutide)	Rapid glucose lowering causing transient microvascular changes	**Observed in SUSTAIN-6**; use caution in patients with pre-existing proliferative DR. Gradual glucose control recommended.
Cardiovascular	Mild tachycardia (↑HR 2–4 bpm)	Direct sinus node stimulation or autonomic modulation	**Clinically insignificant** in most; monitor in heart failure or arrhythmia patients.
Renal	Dehydration-related acute kidney injury	Volume depletion secondary to vomiting or diarrhea	**Rare.** Ensure adequate hydration; monitor renal function during GI side effects.
Injection site/ Dermatologic	Erythema, pruritus, nodules (esp. exenatide ER, Dulaglutide)	Local immune or foreign-body reaction	**Generally mild;** rotate injection sites.
Central Nervous System	Headache, fatigue, dizziness	Central GLP-1 receptor activity or transient metabolic adjustment	**Usually mild;** reassurance sufficient.
Other metabolic	Hypoglycemia (mainly with sulfonylureas or insulin)	Potentiation of glucose-dependent insulin secretion	**Uncommon as monotherapy;** reduce doses of concomitant insulin/SU if used.

5. Emerging pharmacovigilance data

Emerging pharmacovigilance data from national reporting systems have also led to new regulatory actions. The UK Medicines and Healthcare products Regulatory Agency (MHRA) and other authorities have received increased reports of acute pancreatitis and gallbladder events linked to Semaglutide and related agents (MHRA reminds healthcare professionals to). In 2025, the Malaysian National Pharmaceutical Regulatory Agency (NPRA) issued a safety alert regarding aspiration and aspiration pneumonia during anesthesia or deep sedation in patients using GLP-1RAs-an event likely due to delayed gastric emptying (NPRA Accessed). These observations have prompted updates to clinical practice, with anesthesiology societies and diabetes associations now recommending temporary discontinuation of GLP-1RAs before elective procedures to reduce aspiration risk.

Regulatory bodies have also intensified post-marketing surveillance and risk communication for GLP-1RAs as their use expands beyond diabetes into obesity and cardiometabolic prevention. Agencies such as the FDA, EMA, and MHRA are now focusing on ensuring the safe prescribing and sourcing of GLP-1RAs, particularly in response to compounded or counterfeit Semaglutide formulations circulating globally (FDA's Concerns with Unapproved GLP). At the same time, new data continue to support their cardiovascular and metabolic benefits, reinforcing their role in long-term management despite tolerability concerns (Deanfield et al., 2024; EMA finds no link between, 2024). Overall, the 2023-2025 period has been characterized not by the emergence of new safety crises, but by refinement of risk understanding, enhanced pharmacovigilance, and targeted regulatory guidance aimed at optimizing safe and appropriate GLP-1RA use.

Despite these advances, several critical gaps remain. The long-term safety of GLP-1RAs beyond a decade of continuous use is still unclear, particularly concerning rare malignancies and gallbladder or biliary complications. Additionally, more evidence is needed on perioperative management protocols, especially in urgent surgical settings where drug withdrawal is impractical. The safety implications of off-label and compounded formulations, especially in non-diabetic and younger populations, require urgent investigation through prospective real-world studies and international pharmacovigilance harmonization.

Table 3 lists the major trials and studies and regulatory responses related to GLP-1RA adverse effects.

Table 3
Major trials and studies and regulatory responses related to GLP-1RA adverse effects.

Year	Trial (Full Name)	Abbreviation	GLP-1RA Studied	Main Findings on Adverse Effects	Regulatory (FDA/EMA/etc.) Response
2004-2008	Exenatide Clinical Development Program	-	**Exenatide**	Frequent **nausea/vomiting**, early post-marketing reports of **acute pancreatitis**.	**FDA (2008):** Issued safety communication; labeling updated to include **acute pancreatitis warning**.
2008-2010	LEAD Program (*Liraglutide Effect and Action in Diabetes*)	LEAD-1 → LEAD-6	**Liraglutide**	**GI AEs** common; early pancreatitis concerns.	**EMA (2009):** Approved with **risk-management plan** for pancreatitis monitoring. **FDA (2010):** Label included **thyroid C-cell tumor boxed warning** based on rodent data.
2013-2016	ELIXA (*Evaluation of Lixisenatide in Acute Coronary Syndrome*)	ELIXA	**Lixisenatide**	**GI intolerance**; no increased **pancreatitis** risk.	**EMA/FDA (2016):** CV safety confirmed; maintained existing warnings for **GI and pancreatitis** AEs.
2013-2016	LEADER (*Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results*)	LEADER	**Liraglutide**	**No causal link** to pancreatitis or thyroid tumors; **nausea** persistent.	**FDA (2017):** Approved new CV indication; **no new safety restrictions** added.
2015-2018	SUSTAIN-6 (*Trial to Evaluate Cardiovascular and Other Long-term Outcomes with Semaglutide*)	SUSTAIN-6	**Semaglutide (SC)**	**GI intolerance**; mild increase in **retinopathy complications**.	**FDA (2017):** Required **post-marketing retinopathy study**; label updated with **retinopathy warning**.
2015-2019	EXSCEL (*Exenatide Study of Cardiovascular Event Lowering*)	EXSCEL	**Exenatide ER**	**GI AEs** common; **no pancreatitis signal**.	**FDA/EMA (2018):** Confirmed **no increased pancreatic risk**; maintained class-label warnings.
2016-2019	REWIND (*Researching Cardiovascular Events with a Weekly Incretin in Diabetes*)	REWIND	**Dulaglutide**	Similar **GI profile**; no increase in **thyroid tumors** or **pancreatitis**.	**FDA/EMA (2019):** Approved **CV indication**; retained **pancreatitis and thyroid warnings**.
2017-2021	PIONEER Program (*Peptide Innovation for Early Diabetes Treatment*)	PIONEER-1 → PIONEER-10	**Oral Semaglutide**	**GI intolerance** dose-dependent; no new signals.	**FDA (2019):** First oral GLP-1RA approved; labeled with **standard class warnings** (pancreatitis, thyroid C-cell tumors).
2018-2021	AWARD Program (*Assessment of Weekly AdministRation of Dulaglutide*)	AWARD-1 → AWARD-11	**Dulaglutide**	**GI intolerance** frequent; rare **gallbladder events** reported.	**EMA (2020):** Updated product information to include **biliary/gallbladder disease** as potential risk.
2021-2023	STEP Program (*Semaglutide Treatment Effect in People with Obesity*)	STEP-1 → STEP-8	**Semaglutide (2.4 mg)**	**High GI AE rates**, including **nausea/vomiting**; **gallbladder disorders** noted.	**FDA (2021):** Approved for obesity (Wegovy); labeling includes **gastrointestinal and gallbladder warnings**.
2023	SELECT (*Semaglutide Effects on Heart Disease in Overweight or Obese People without Diabetes*)	SELECT	**Semaglutide**	**GI intolerance** and **biliary events** most frequent AEs; confirmed CV benefit.	**FDA (2024, anticipated):** Considering **expanded CV indication**; no new safety issues flagged.

6. Pharmaceutical, regulatory, clinical and public health implications

Since their introduction, GLP-1 receptor agonists (GLP-1RAs) have profoundly reshaped the antidiabetic and obesity treatment landscape, with global sales surpassing USD 30 billion in 2024 (Zhao et al., 2021). Novo Nordisk and Eli Lilly dominate this market through semaglutide and tirzepatide, respectively, and the dual indications for diabetes and obesity have expanded market share while stimulating innovation in extended-release formulations and oral delivery systems (Melson et al., 2025).Prescription patterns reflect a shift from daily to once-weekly dosing, enhancing adherence and patient convenience (Ukhanova et al., 2025).

The growing off-label use of GLP-1RAs for weight loss in non-diabetic populations has raised safety concerns, including potential misuse outside medical supervision (FDA's Concerns with Unapproved GLP). High demand has led to intermittent global shortages, prompting regulatory interventions to prioritize access for patients with diabetes. This rapid adoption also carries substantial pharmacoeconomic implications, as costs often exceed USD 1000 per month, limiting access in low- and middle-income countries (LMICs) (The WHO Global Diabetes Compact). Consequently, payers and public health systems are reevaluating reimbursement strategies and cost-effectiveness thresholds.

From a regulatory perspective, agencies emphasize ongoing post-marketing surveillance, including periodic safety update reports (PSURs) and the generation of real-world evidence to monitor adverse events and emerging safety concerns. Knowledge of GLP-1RA adverse effects is critical for guiding clinical practice, informing patient counseling, and shaping regulatory decisions. Clinicians must weigh benefits against risks, particularly gastrointestinal intolerance, gallbladder disease, and rare systemic effects, while considering individual patient characteristics and comorbidities.

Overall, the combined impact on the pharmaceutical industry, prescribing trends, regulatory oversight, and clinical practice underscores the importance of a holistic understanding of GLP-1RA use. This integrated perspective ensures safe, effective, and evidence-informed deployment of these agents across diverse patient populations.

7. Recommendations for safer use

Clinical and regulatory best practices emphasize a comprehensive, patient-centered approach to the safe use of GLP-1RAs. Before initiating therapy, a pre-treatment assessment should be conducted to evaluate patients for a history of pancreatitis, gallbladder disease, thyroid disorders, or renal impairment, as these conditions may increase the risk of adverse events or necessitate closer monitoring. Gradual dose escalation is essential-starting at the lowest effective dose and titrating slowly-to minimize gastrointestinal intolerance, which remains the most common cause of discontinuation. Clinicians should also monitor patients' psychological status, particularly for mood changes or suicidal ideation, given emerging reports linking GLP-1RA use to neuropsychiatric effects in susceptible individuals.

Equally important is the promotion of pharmacovigilance, encouraging both clinicians and patients to report any suspected adverse



Fig. 2. GLP-1 receptor agonists in Obesity
Safety profile and clinical implications.

reactions promptly to enhance real-world safety data and guide regulatory action. From a policy perspective, ensuring equitable access remains critical; affordability, availability, and responsible prescribing frameworks should be prioritized to prevent inappropriate or inequitable use. Continuous education and communication among healthcare professionals are vital to keep pace with rapidly evolving evidence and updated safety recommendations, fostering informed, evidence-based prescribing of GLP-1RAs.

Fig. 2 presents an overview of GLP-1 receptor agonist safety, highlighting common dose-dependent gastrointestinal effects, rarer serious adverse events, underlying mechanisms, and key regulatory milestones.

8. Conclusion and future research directions

GLP-1 receptor agonists have transformed the therapeutic landscape of diabetes and obesity, providing robust glycemic control, substantial weight loss, and cardiometabolic protection. However, as their clinical reach expands, recognition of their adverse-effect spectrum has become increasingly important. The most common reactions-gastrointestinal discomfort, gallbladder events, and mild heart-rate increase-are generally manageable, whereas rare events such as pancreatitis or psychiatric symptoms warrant vigilance.

The evolving evidence indicates that the benefit–risk balance of GLP-1RAs remains favorable when appropriately prescribed and monitored. Future efforts should focus on optimizing pharmacovigilance systems, understanding patient-specific predictors of intolerance, and promoting equitable access to ensure maximal therapeutic benefit with minimal harm.

Future research directions should focus on several critical domains to address existing knowledge gaps regarding GLP-1RAs. Mechanistic studies are needed to clarify the cellular and molecular pathways linking GLP-1 signaling to gastrointestinal motility, pancreatic stress, and mood regulation, which remain incompletely understood despite extensive clinical data. Large-scale long-term cohort studies are also essential to evaluate the cumulative safety of GLP-1RAs in individuals with over a decade of exposure, particularly concerning cancer risk and other delayed adverse outcomes.

In parallel, comparative real-world analyses should be conducted to determine whether adverse-event patterns differ between monotherapy with GLP-1RAs and dual-agonist therapies (such as GLP-1/GIP receptor agonists), given their expanding clinical use. Advances in digital pharmacovigilance-including predictive algorithms that integrate electronic health records, patient-reported data, and wearable device metrics-could substantially enhance early detection of rare or emerging safety signals. Moreover, health-equity research is vital to explore disparities in access, affordability, and adverse-event reporting across diverse populations and regions. Finally, public communication studies should examine how social media narratives and direct-to-consumer marketing influence patient perceptions of benefit and risk, shaping adherence and real-world safety outcomes.

Additionally, future research should specifically investigate the safety of GLP-1RAs during and after pregnancy. This includes women using these agents to improve fertility as well as those with pregnancies complicated by pregestational or gestational diabetes, in order to better understand potential impacts on maternal and fetal outcomes and the role of GLP-1RAs in preventing or managing future metabolic disease in both mother and child.

CRediT authorship contribution statement

Riad Mohammed Abdelrahman: Writing – review & editing, Methodology, Conceptualization. **Taha Hussein Musa:** Writing – review & editing, Conceptualization. **Ismail Adam Arbab:** Writing – original draft, Methodology. **Eltieb Omer Ahmed:** Writing – review & editing, Conceptualization. **Sahar Ibrahim Gasmallah:** Writing – original draft, Methodology. **Mohammed Jalal:** Methodology, Conceptualization. **Chiamaka Linda Mgbechidinma:** Writing – review & editing, Methodology, Conceptualization. **Wafaa Ramadan Ahmed:** Writing – review & editing, Conceptualization.

Availability of supporting data

The data supporting the findings of this study are available on the websites mentioned in the References section.

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Not applicable.

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During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using that, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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